



NITI Aayog



SECTOR AT A GLANCE 08

River Rejuvenation in India

Water in India

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RIVER REJUVENATION IN INDIA

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

River rejuvenation refers to the sustainable revitalisation of river ecosystems for economic development, improved livelihoods, and river health. This concept gained momentum around the 2010s as rivers were acknowledged as crucial for economic growth and achieving several UN Sustainable Development Goals.¹ These include Goal 6 (clean water and sanitation), reflecting the importance of clean rivers for public health and hygiene; Goal 8 (decent work and economic growth), highlighting the role of rivers in supporting livelihoods and local economies; Goal 11 (sustainable cities and communities), underscoring the role of rivers in urban planning and development; Goal 13 (climate action), emphasising the role of rivers in climate regulation; and Goal 14 (life below water), recognising the biodiversity in river ecosystems.²

However, rapid unplanned urbanisation and industrialisation have heavily affected rivers in India. Nearly 46% of the rivers monitored (279 out of 603) were identified as polluted in 2022.³ Similarly, depleting river flows threaten human activity and health, aquatic ecosystems, agricultural productivity, and economic growth.

With policy support, investments, and monitoring, river rejuvenation can enhance water availability, improve health and livelihoods, and drive sustainable development across the country.

Given these opportunities and challenges, riverfront development, river cleaning, setting emission standards and overall river quality improvement have been major policy focus areas. Several ongoing and recent government initiatives such as the Namami Gange program for the Ganga basin rivers and the National River Conservation Plan (NRCP) for other rivers focus on this. These efforts, along with several state- and local-level policies and initiatives, seek to ensure that India can revive the quality of its rivers and attain water security.

Figure 1: Key River Rejuvenation Statistics in India

Estimated benefits from the rejuvenation of 13 major rivers

1900 million m³ yr⁻¹

Groundwater recharge

50 million tonnes of CO₂e

Additional carbon sequestration in ten years

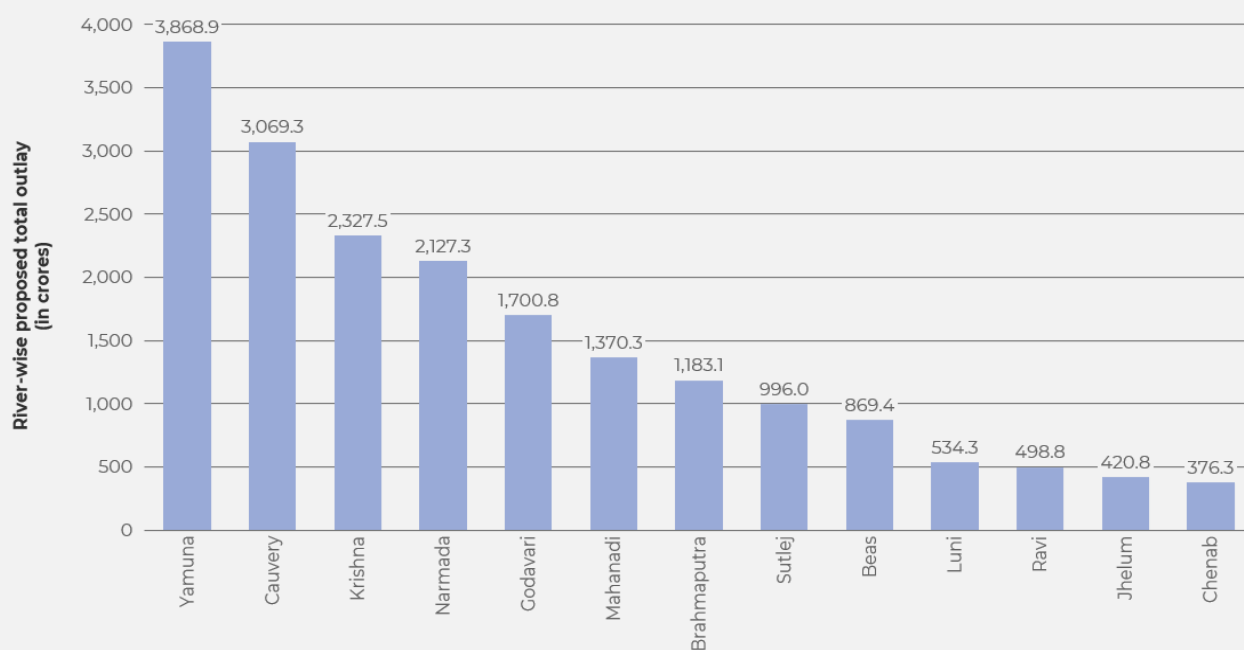
Source: Indian Council of Forestry Research and Education, Overview of Detailed Project Reports for Rejuvenation of Major Indian Rivers through Forestry Interventions, 2022

¹ UNDESA, International Decade for Action 2005-2015, accessed 2023

² UN, Sustainable Development Goals, accessed 2023

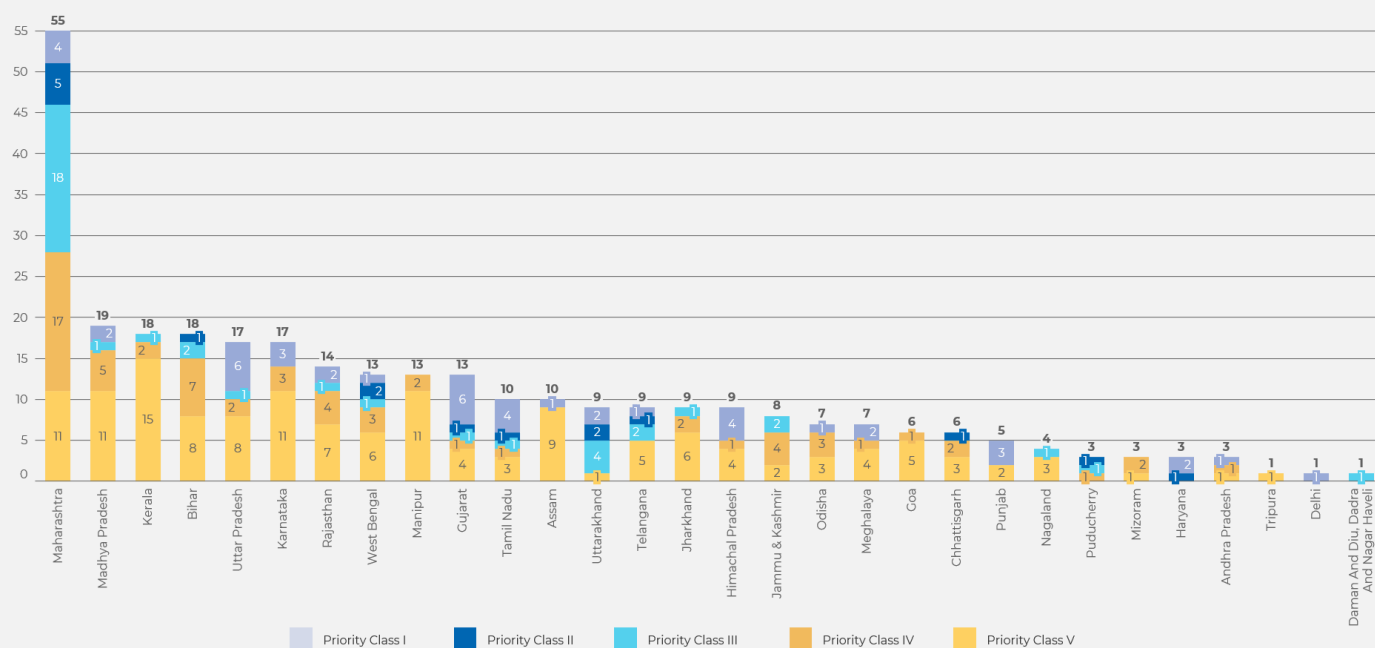
³ Ministry of Environment, Forest, and Climate Change, Analysis of water quality of rivers by Central Pollution Control Board, 2023

Figure 2: Budget outlay for rejuvenation and conservation of thirteen major Indian rivers



Source: Indian Council of Forestry Research and Education, Overview of Detailed Project Reports for Rejuvenation of Major Indian Rivers through Forestry Interventions, 2022

Figure 3: State-wise & priority-wise number of polluted river stretches (2023)



Source: Ministry of Environment, Forest and Climate Change, Analysis of water quality of rivers by Central Pollution Control Board, 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective river rejuvenation practices. These revolve primarily around the potential for improved water quality, public health, biodiversity protection and economic growth.



Improved water quality and availability:

India's rivers receive a significant volume of sewage and untreated waste daily.⁴ For instance, 72% of the wastewater remains untreated and may be disposed of in rivers, lakes, or groundwater.⁵ Reducing pollution through treatment, waste management, and limiting chemical runoff can increase water suitable for irrigation, industry, and domestic use.⁶ Managing water extraction, especially during dry periods, can also raise river flow, groundwater levels, and water security.



Better public health:

Untreated sewage and industrial waste from hundreds of polluted river stretches drive significant healthcare challenges for India.⁷ The health costs relating to water pollution are estimated at about INR 470-610 billion (\$6.7-8.7 billion per year) – most associated with diarrheal mortality and morbidity of children under five and other morbidities.⁸ Preventing such waste from entering rivers will reduce disease burdens from water contamination and make more water safe for public use. This can decrease health costs significantly over time.⁹



Improved economic growth:

Improvements to water quality, water availability, and biodiversity along rivers can boost tourism, river transportation, agriculture production, and fisheries, generating revenue and jobs, especially for rural communities.¹⁰ Improving river water quality could also drive improvements in economic growth in downstream regions.¹¹



Greater biodiversity conservation:

Protecting river habitats and managing water flows sustainably can support more diverse and abundant fish and wildlife populations, including endangered species like the Ganges dolphin.¹² This enhances biodiversity and opportunities for eco-tourism which the government aims to promote along riverbanks, and potentially in estuaries.¹³

Challenges

This section highlights some of the major challenges and constraints facing river rejuvenation in India. Efforts to resolve such challenges will ensure more efficient river water extraction, reduction of pollution, and scientific and sustainable sand mining.

⁴ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁵ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁶ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁷ NITI Aayog, Urban Wastewater Scenario in India, 2022

⁸ WEF, Water pollution is killing millions of Indians. Here's how technology and reliable data can change that, 2019

⁹ WEF, Water pollution is killing millions of Indians. Here's how technology and reliable data can change that, 2019

¹⁰ UNEP, Four reasons to protect rivers, 2022

¹¹ WEF, Water pollution is killing millions of Indians. Here's how technology and reliable data can change that, 2019

¹² NMCG, The Conservation Action Plan For The Gangetic Dolphin 2010-2020, 2010

¹³ NITI Aayog, Urban Wastewater Scenario in India, 2022



Water scarcity and over-extraction:

The pressure on freshwater resources leads to diminishing flows in some river systems. This is mainly due to water withdrawal for irrigation, which is often to the tune of 70-80% of the lean season flows, with some proportion for domestic and industrial use.¹⁴ Consequently, rivers in India run dry for months due to over-extraction.¹⁵



Encroachment and pollution in rivers:

River encroachment for agriculture, industry, or residences or construction-related blockages of stormwater drains and associated inflows into the river destroy habitats, reduce the capacity of rivers to carry flood waters, destroy natural wetlands, and worsen flood risk.¹⁶ Further, a 2022 report by the Central Pollution Control Board indicates that out of the 351 polluted river stretches, 13% of rivers are highly polluted.¹⁷ Pollution and barriers like poorly planned dams threaten fish and wildlife in river ecosystems.¹⁸ For instance, Brahmaputra River dolphins, gharials in Yamuna, and numerous fish species are now endangered.¹⁹



Illegal and unscientific sand mining:

While some states have banned sand mining under Section 23C of the Mines and Minerals (Development & Regulation) Act, 1957 (MMDR Act), continued illegal and unscientific sand mining severely threatens river ecology. Excessive sand mining can change the riverbed, force the river to change its course, erode riverbanks, cause flooding, and destroys habitat of aquatic animals as well as micro-organisms.²⁰

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14 MDPI, Securing Flows in the River Systems through Irrigation Water Use Efficiency—A Case Study from Karula River in the Ganga River System, 2022

15 MDPI, Securing Flows in the River Systems through Irrigation Water Use Efficiency—A Case Study from Karula River in the Ganga River System, 2022

16 MoHUA, River Centric Urban Planning Guidelines, 2021

17 CPCB, Polluted River Stretches For Restoration Of Water Quality, 2022

18 CWE Journal, Current Status of the Endangered Ganges River Dolphin (Platanista Gangetica), the Aquatic Megafauna in the Brahmaputra River System, 2021

19 CWE Journal, Current Status of the Endangered Ganges River Dolphin (Platanista Gangetica), the Aquatic Megafauna in the Brahmaputra River System, 2021

20 UN Environment Programme, Sand and Sustainability: 10 Strategic Recommendations to Avert a Crisis, 2022

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee river rejuvenation.



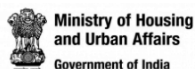
Ministry of Jal Shakti²¹

The Ministry aims to effectively manage water as a national resource, including river water. It houses the Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) which is responsible for establishing policy guidelines and programmes for the management, development, and conservation of river water.



Ministry of Environment, Forest, and Climate Change

The Ministry is the nodal agency for the planning, promotion, co-ordination and overseeing the implementation of India's environmental and forestry policies and programmes. It releases Detailed Project Reports (DPRs) on river rejuvenation and aims to prevent pollution of rivers and improve water quality.



Ministry of Housing and Urban Affairs

The ministry oversees the formulation and administration of the rules and regulations and laws pertaining to urban development in India. It creates master plans and guidelines for river conservation in urban areas.



Central Pollution Control Board²²

The board is a statutory organization under the Ministry of Environment, Forest and Climate Change. It monitors water quality across lakes, ponds, wells and rivers, and identifies polluted river stretches.

Major policies and programs

This section highlights existing policies related to river rejuvenation and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Namami Gange²³

This is a flagship scheme by the Ministry of Jal Shakti to rejuvenate river Ganga and associated tributaries. The total program cost is INR 20,000 crores, and projects include

²¹ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

²² Ministry of Environment, Forest and Climate Change, Analysis of water quality of rivers by Central Pollution Control Board, 2023

²³ Namami Gange, Namami Gange Programme, 2020

creating sewage treatment plants, riverfront development, biodiversity projects, and river surface cleaning projects.²⁴



National River Conservation Plan²⁵

This is a key program implemented by the Ministry of Jal Shakti to abate pollution and improve water quality in major rivers excluding those in the Ganga basin, by providing financial and technical support to States and UTs.



Environment Protection Act, 1986²⁶

This Act allows the government to set discharge and emission standards for various pollutants, thereby controlling the quality of water in Indian rivers.



Wetlands (Conservation and Management) Rules of 2017²⁷

Wetlands act as natural sponges, absorbing excess rainwater and slowly releasing it into river systems, thereby regulating water flow and reducing flooding risk. These rules allow for more localised, effective conservation strategies that can directly enhance these benefits to river systems.²⁸

Way forward

India has commenced an ambitious program for rejuvenating its rivers under schemes like Namami Ganga, NRCP, and DPRs worth INR 19,000 were released in 2022 by the Ministry of Environment, Forest & Climate Change to support the rejuvenation of 13 major rivers. The focus includes *aviral dhara* (continuous flow), *nirmal dhara* (unpolluted flow), and ecological restoration.

These developments present a significant opportunity to ensure that India's rivers get revitalised and support critically important human, ecosystem, and economic objectives.

Some principles that could ensure greater success in these programs include:



Strengthening monitoring and

- Establish comprehensive monitoring frameworks for objective measurement of ecological, economic, and social outcomes at the river system and state levels against predetermined, clear, and measurable targets to track progress and identify areas of improvement.

²⁴ Namami Gange, Namami Gange Programme, 2020

²⁵ Ministry of Jal Shakti, National River Conservation Plan, accessed 2023

²⁶ Ministry of Environment, Forest and Climate Change, The Environment (Protection) Act, 1986

²⁷ Ministry of Environment, Forest and Climate Change, Wetlands (Conservation and Management Rules) 2017

²⁸ Ministry of Environment, Forest and Climate Change, Wetlands (Conservation and Management Rules) 2017

evaluation for river systems	
 Enhancing real-time tracking of water flows	• Mandate continuous real-time tracking of environmental water flows, considering seasonal variations through sensor-based systems for frequent measurement of key contaminants to enable informed and timely interventions.
 Building sustainable operational models for SETPs	• Develop sustainable operational models for existing and planned sewage effluent treatment plants via innovative financing tools such as performance-based financing instruments to ensure long-term functionality and maintenance. • Promote community buy-in through awareness programs and establish industry tariffs for waste treatment.
 Promoting collaborative expertise and knowledge sharing	• Recognise and manage the interconnectedness and complexities of river systems by forging alliances between research institutions and civil society organizations for (i) deep technical expertise (ii) knowledge of local and understanding of communities and their needs for site-specific effective solutions.

Recommended readings

UN Environment Programme. (2022). *Four reasons to protect rivers*.

<https://www.unep.org/news-and-stories/story/four-reasons-protect-rivers>

NITI Aayog. (2022). *Urban Wastewater Scenario in India*.

https://www.niti.gov.in/sites/default/files/2022-09/Waste-Water-A4_20092022.pdf

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Ministry of Jal Shakti. (2020). *Namami Gange Programme – At a Glance*.

[https://nmcg.nic.in/pdf/NGP-At%20a%20Glance%20\(Final%20Version%20Printed\).pdf](https://nmcg.nic.in/pdf/NGP-At%20a%20Glance%20(Final%20Version%20Printed).pdf)

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